Lab 5 - Part 2

RoboLink

Course: RoboLink: Build and Drive Your Smartphone-Controlled Robot
Program: Mini Course Program 2026, Carleton University
Instructors: Dr. Mahmoud Sayed and Mahmoud Yassin

Lab Mission
In this lab, you will use the RoboLink Android app to drive a two-motor robot. The app sends wireless BLE commands to the Arduino Nano. The Arduino reads the command and controls the L9110 2-channel motor driver so the robot can move forward, move backward, turn left, turn right, and stop.

Group Member Names ______________________________

Date ______________________________

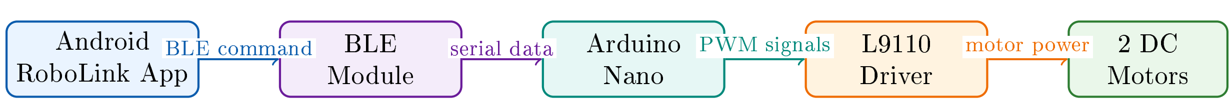
1. Learning Goals

By the end of this lab, you should be able to:

- Control a robot from an Android phone using BLE commands.
- Connect a BLE module and an L9110 motor driver to an Arduino Nano ATmega168.
- Explain how one command number becomes a robot movement.
- Use the Serial Monitor to debug wireless robot commands.
- Tune motor speed values so the robot drives more predictably.
- Troubleshoot common wiring, upload, connection, and movement problems.

2. Big Idea

A robot command travels through a chain.
The phone does not power the motors directly. It only sends a small command. The Arduino decides what the command means, and the motor driver supplies the motor current needed to move the robot.



3. Important Safety and Setup Notes

Keep the Wheels Off the Table First
For the first test, lift the robot so the wheels can spin freely. This prevents the robot from driving off the table while you are checking the wiring and app connection.

Use Motor Power Correctly
Do not power the motors only from the Arduino Nano 5V pin or USB. Use the motor battery or external motor power supply for the L9110 VCC pin. Connect Arduino GND, BLE module GND, L9110 GND, and battery GND together.

Special BLE Edition
The module used in this lab looks like an HC-05 module, but it is a special edition that works with **BLE**, not classic Bluetooth SPP. Use the RoboLink Android app provided by your instructor.

4. Materials

Arduino Nano ATmega168	Reads commands and controls the robot.
Special BLE HC-05-style module	Receives wireless commands from the Android app.
L9110 2-channel motor driver	Drives the left and right DC motors.
2 DC motors and robot chassis	Creates robot movement.
Android phone	Runs the RoboLink app. iPhone is not supported in this lab.
USB cable	Uploads code and opens the Serial Monitor.
Motor battery or external motor supply	Powers the motors through the L9110 driver.
Jumper wires and voltage divider resistors	Connect the modules safely.

5. Command Map

The RoboLink app sends one command number when a direction button is pressed. When the button is released, the app sends `1` so the robot stops.

Command	Robot Action	When It Is Sent
1	Stop	Button released or Stop pressed
2	Move forward	Forward pressed
3	Move backward	Backward pressed
4	Turn left	Left pressed
5	Turn right	Right pressed

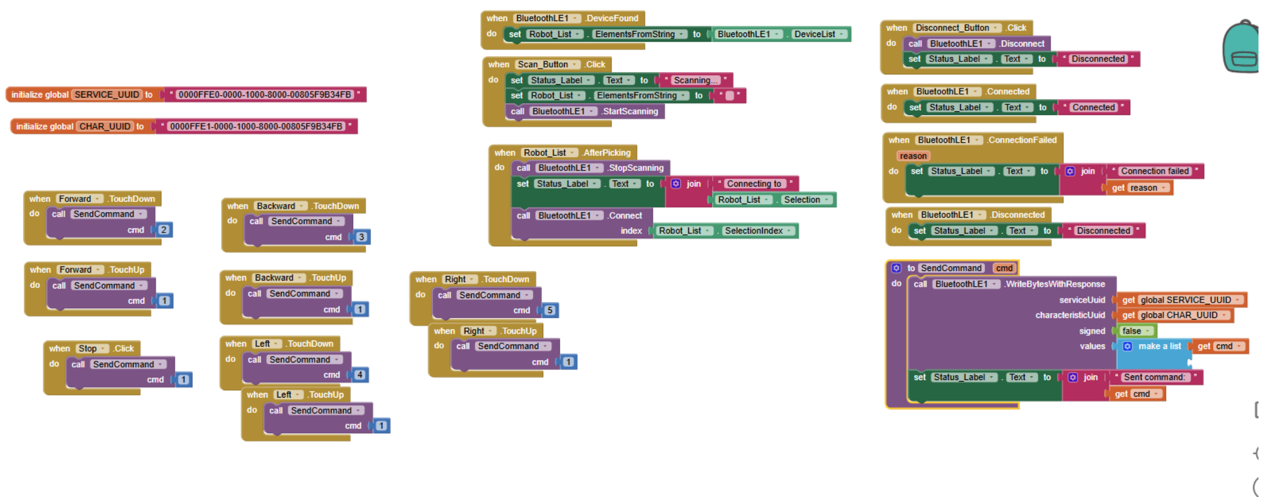
6. Install and Prepare the Android App

Android Only
The mobile app for this lab works on **Android phones only**. It is not designed for iPhone.

Before Scanning for the Robot
Turn on **Bluetooth** and **Location/GPS**. Also give the RoboLink app all requested Android permissions for connectivity and location, such as Nearby Devices, Bluetooth, and Location permissions.

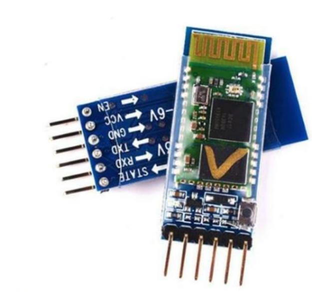
Install the app from this link:

https://github.com/MYMahmoudYassin/MPC-2026-RoboLink/raw/main/Day5_RoboLink_Smartphone_Control/Labs/RoboLink%20APP/Robotak%20Controller.apk



MIT App Inventor blocks. The app scans for a BLE module, connects to it, and sends command values when the robot buttons are used.

7. Wiring the BLE Module

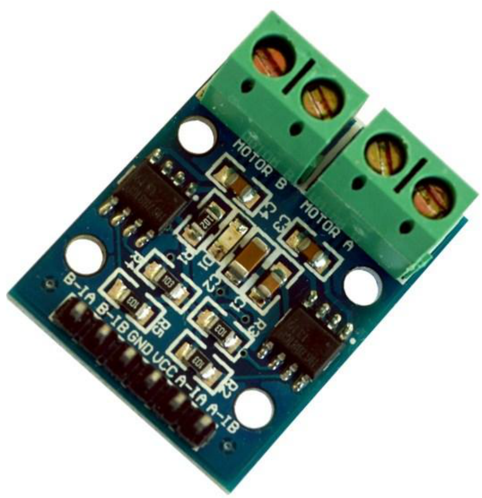


BLE module. The module pins are labelled EN, VCC, GND, TXD, RXD, and STATE.

Protect the Module RXD Pin
Arduino D11 is a 5V signal. Many BLE module RXD pins expect about 3.3V. Use a voltage divider between Arduino D11 and module RXD.

BLE Module Pin	Arduino Nano Pin	Notes
TXD	D10	Module transmits to Arduino receive pin.
RXD	D11	Use a voltage divider before RXD.
GND	GND	Must share ground with the Arduino and motor supply.
VCC	5V or 3.3V	Use the correct power for your module board.
EN	Not used	Leave unconnected for this lab.
STATE	Not used	Leave unconnected for this lab.

8. Wiring the L9110 Motor Driver



L9110 2-channel motor driver. The green screw terminals connect to Motor A and Motor B.

L9110 Pin	Arduino Nano Pin	Purpose
A-IA	D3	Left motor forward PWM signal.
A-IB	D5	Left motor backward PWM signal.
B-IA	D6	Right motor forward PWM signal.
B-IB	D9	Right motor backward PWM signal.
GND	GND	Connect to Arduino GND and battery GND.
VCC	Motor power	External motor power, usually 2.5V to 12V.

Before Powering the Motors
Ask your teacher to check the BLE wiring, L9110 wiring, battery polarity, and shared ground before turning on motor power.

9. Arduino IDE Setup

1. Open the Arduino IDE or PlatformIO project.
2. Open the Lab 5 Part 2 RoboLink code.
3. Select **Tools** → **Board** → **Arduino AVR Boards** → **Arduino Nano**.
4. Select **Tools** → **Processor** → **ATmega168**.
5. Select the correct **Tools** → **Port**.
6. Upload the code.
7. Open **Tools** → **Serial Monitor**.
8. Set the baud rate to **9600**.

10. Experiment Procedure

Part A: USB Test First

Before using the app, test the code from the Serial Monitor.

1. Keep the robot wheels lifted off the table.
2. Open the Serial Monitor at 9600 baud.
3. Send `1`, `2`, `3`, `4`, and `5`.
4. Watch the wheels and read the printed messages.

Command	Expected Movement	What Actually Happened?
1	Stop	
2	Forward	
3	Backward	
4	Turn left slowly	
5	Turn right slowly	

Part B: Connect the RoboLink App

1. Turn on Bluetooth and Location/GPS on the Android phone.
2. Check that the app permissions are allowed.
3. Open the RoboLink app.
4. Scan for the BLE module.
5. Select the robot BLE module and wait for connection.
6. Keep the Serial Monitor open so you can see the commands arriving.

Connection observation: What name appeared for your BLE module?

Part C: Drive the Robot from the App

1. Press and hold Forward. Release the button and check that the robot stops.
2. Press and hold Backward. Release the button and check that the robot stops.
3. Press and hold Left. Release the button and check that the robot stops.
4. Press and hold Right. Release the button and check that the robot stops.
5. If the robot moves safely while lifted, place it on the floor and test again.

Observation: Did the robot stop when you released each app button?

Part D: Tune the Movement

The code has three important speed values:

```
const int LEFT_MOTOR_SPEED = 200;
const int RIGHT_MOTOR_SPEED = 200;
const int TURN_SPEED = 140;
```

- If the robot curves left while going forward, the left motor may be slower.
- If the robot curves right while going forward, the right motor may be slower.
- If turning is too weak, increase `TURN_SPEED` a little.
- If turning is too fast, decrease `TURN_SPEED` a little.

Your tuning notes:

Part E: Mini Driving Challenge

Use the app to drive your robot through a simple path chosen by your teacher. Drive slowly and stop if the robot leaves the course.

What strategy helped you control the robot smoothly?

11. What Is Happening in the Code?

`SoftwareSerial BT(10, 11)`	Creates a serial connection to the BLE module using Arduino pins D10 and D11.
`BT.available()`	Checks whether the phone app sent a command.
`BT.read()`	Reads one command byte from the BLE module.
`normalizeCommand()`	Allows both app commands and Serial Monitor test commands to work.
`handleCommand()`	Chooses stop, forward, backward, left, or right.
`analogWrite()`	Sends PWM values to control motor speed.
`stopMotors()`	Sets all four L9110 input pins LOW so the robot stops.
`moveTo()`	Adds a short pause when changing direction.

Why are some motor pins LOW?
Each L9110 motor channel uses two input pins. To spin one direction, one pin receives PWM and the other pin is LOW. To spin the opposite direction, the pins switch roles. When stopping, all four driver input pins are set LOW.

12. Expected Problems and Fixes

Problem	Possible Cause	What To Try
Upload fails	Wrong board, processor, or port	Select Arduino Nano, ATmega168, and the correct port.
App cannot find module	Bluetooth off, Location/GPS off, permissions missing, or BLE module not powered	Turn on Bluetooth and Location/GPS, allow app permissions, and check module power.
Serial Monitor shows no app commands	TXD/RXD wiring problem or wrong baud rate	Set Serial Monitor to 9600. Check TXD to D10 and D11 to RXD through the voltage divider.
Nothing moves	Motor battery missing, shared ground missing, or L9110 wiring error	Check motor power, GND connections, D3, D5, D6, and D9.
Robot moves when button is released	Stop command is not arriving or app connection dropped	Check Serial Monitor for command 1 and reconnect the app.
Robot drives backward on Forward	Motor wires are reversed	Swap the two wires on the motor terminals or adjust the movement function.
Left and right are reversed	Motor A and Motor B are swapped, or one motor is reversed	Check which motor is connected to Motor A and Motor B.
One motor is slower	DC motors are not perfectly matched	Tune `LEFT_MOTOR_SPEED` and `RIGHT_MOTOR_SPEED`.
Turns are too fast or too slow	`TURN_SPEED` needs tuning	Adjust `TURN_SPEED` in small steps.
Robot resets when motors start	Motor supply cannot provide enough current	Use a stronger motor battery or reduce speed.

13. Reflection Questions

Answer in full sentences.

1. Why does the app send a stop command when you release a direction button?
2. Why should we test the robot with the wheels lifted before driving it on the floor?
3. What is the job of the Arduino in this robot system?
4. What is the job of the L9110 motor driver?
5. Why do the Arduino, BLE module, L9110 driver, and motor battery need a shared GND?

6. If the robot does not drive straight, what evidence would help you decide which motor speed to change?

7. What is one improvement you would make to the RoboLink robot or app?

14. Exit Ticket

In one or two sentences: Explain how pressing a button on a phone can make a robot move.

15. Teacher Check

Teacher Checklist

- ☐ **BLE module wiring checked** Notes: ____________
- ☐ **L9110 motor wiring checked** Notes: ____________
- ☐ **Code uploaded** Notes: ____________
- ☐ **Android app connected to BLE module** Notes: ____________
- ☐ **Robot responds to app commands** Notes: ____________
- ☐ **Robot stops when app button is released** Notes: ____________
- ☐ **Students completed observations** Notes: ____________
- ☐ **Students answered reflection questions** Notes: ____________